

Definition 2.2 (Prefix Probability). $\pi_l^<(\mathbf{y}_{<l}) := \prod_{j=1}^{l-1} \pi(y_j | \mathbf{x}, \mathbf{y}_{<j})$, which represents the cumulative probability of the model generating the prefix $\mathbf{y}_{<l}$.

Definition 2.3 (Future Alignment Potential). $q_{>l}^+(\mathbf{y}_l) := P_\pi((\mathbf{x}, \mathbf{y}) \in \mathcal{A} | \mathbf{x}, \mathbf{y}_{<l}, \mathbf{y}_l)$, which denotes the conditional probability that the full completion will eventually belong to the aligned set $\mathcal{A}$, given the current prefix and the choice of the l -th token.

With these two auxiliary variables, we arrive at the following key lemma that bridges the gap between global alignment and local token dynamics.

Lemma 2.4 (Token-level Decomposition). *Consider a small perturbation in the model parameters $\theta \rightarrow \theta + \Delta\theta$, which induces a change $\Delta\pi(\mathbf{y}_l | \mathbf{x}, \mathbf{y}_{<l})$ in the token-level policy. To the first order of $\Delta\pi$, the change in the alignment score ΔS can be decomposed as follows:*

$$\Delta S = \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x})} \sum_{l=1}^{L_y} \sum_{\mathbf{y}_{<l} \in \mathcal{V}^{l-1}} \pi_l^<(\mathbf{y}_{<l}) \cdot \sum_{i \in \mathcal{V}} q_{>l}^+(i) \cdot \Delta\pi(\mathbf{y}_l = i | \mathbf{x}, \mathbf{y}_{<l}) + \mathcal{O}(\|\Delta\pi\|^2). \quad (3)$$

The Lemma reveals that the contribution of a token-level update is “gated” by the prefix probability $\pi_l^<$. Mathematically, earlier tokens in a sequence typically correspond to larger $\pi_l^<$ values, as the cumulative product has fewer terms and thus higher probability density. This result resonates with the phenomenon of shallow alignment [19], where the model’s alignment behavior is predominantly influenced by initial tokens.

2.3 Logit-Space Geometry of Alignment Dynamics

Building on the token-level decomposition above, we now connect alignment dynamics to perturbations in logit space. At each autoregressive state $(\mathbf{x}, \mathbf{y}_{<l})$, let π_l denote the next-token distribution induced by the current policy, $\mathbf{z}_l$ the corresponding logits, and $\mathbf{J}_l := \text{Diag}(\pi_l) - \pi_l \pi_l^\top$ the Jacobian of the softmax map. We also write $\pi_l^<(\mathbf{y}_{<l})$ for the prefix probability introduced above, and let $q_{>l}^+(i)$ denote the probability that the final completion belongs to the aligned set $\mathcal{A}$ after choosing token i at step l .

Definition 2.5 (Alignment probability after a prefix). For a fixed prefix state $(\mathbf{x}, \mathbf{y}_{<l})$, define

$$\pi_{S,l}^+ := P_\pi((\mathbf{x}, \mathbf{y}) \in \mathcal{A} | \mathbf{x}, \mathbf{y}_{<l}), \quad \pi_{S,l}^- := 1 - \pi_{S,l}^+.$$

Here, $\pi_{S,l}^+$ is the probability that the model will eventually generate an aligned completion given the current prefix, while $\pi_{S,l}^-$ is the complementary probability of producing a non-aligned completion.

Definition 2.6 (Aligned and non-aligned conditional token distributions). For the same prefix state $(\mathbf{x}, \mathbf{y}_{<l})$, define

$$\tilde{\pi}_l^+(i) := P_\pi(\mathbf{y}_l = i | \mathbf{x}, \mathbf{y}_{<l}, (\mathbf{x}, \mathbf{y}) \in \mathcal{A}), \quad \tilde{\pi}_l^-(i) := P_\pi(\mathbf{y}_l = i | \mathbf{x}, \mathbf{y}_{<l}, (\mathbf{x}, \mathbf{y}) \notin \mathcal{A}).$$

That is, $\tilde{\pi}_l^+$ and $\tilde{\pi}_l^-$ are the token distributions conditioned on the completion being aligned or non-aligned, respectively.

For a small logit perturbation $\Delta\mathbf{z}_l$, the induced probability change satisfies

$$\Delta\pi_l = \mathbf{J}_l \Delta\mathbf{z}_l + \mathcal{O}(\|\Delta\mathbf{z}_l\|^2).$$

To relate the future alignment potential to the aligned and non-aligned conditional distributions, we view $q_{>l}^+(\mathbf{y}_l)$ as an alignment-conditioned weight and apply Bayes’ rule to rewrite the corresponding local term:

$$\mathbf{q}_{>l}^{+\top} \mathbf{J}_l = \pi_{S,l}^+ \pi_{S,l}^- (\tilde{\pi}_l^+ - \tilde{\pi}_l^-)^\top, \quad (4)$$

Where the bold notation $\mathbf{q}_{>l}^+$ is a vector defined over the vocabulary $\mathcal{V}$. This identity converts the future-dependent quantity $\mathbf{q}_{>l}^+$ into a local contrast between two conditional distributions, one induced by aligned completions and the other by non-aligned completions.

Theorem 2.7 (Logit-space form of alignment dynamics). *Suppose the next-token distribution at each prefix state $(\mathbf{x}, \mathbf{y}_{<l})$ is perturbed by a small logit update $\Delta\mathbf{z}_l$. Then the first-order change in the alignment score satisfies*

$$\Delta S = \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x})} \sum_{l=1}^{L_y} \sum_{\mathbf{y}_{<l} \in \mathcal{V}^{l-1}} \pi_l^<(\mathbf{y}_{<l}) \cdot \pi_{S,l}^+ \pi_{S,l}^- (\tilde{\pi}_l^+ - \tilde{\pi}_l^-)^\top \Delta\mathbf{z}_l + \mathcal{O}(\|\Delta\mathbf{z}\|^2). \quad (5)$$

Equivalently, the alignment score changes according to the projection of the logit update onto the direction that **separates aligned and non-aligned conditional token distributions**, gated by the prefix probability $\pi_l^<(\mathbf{y}_{<l})$ and the **alignment uncertainty factor** $\pi_{S,l}^+ \pi_{S,l}^-$.

2.4 Evolutionary Dynamics of Alignment during Fine-tuning

To characterize the trajectory of the alignment score $S(\theta)$ during optimization, we bridge the geometric analysis of the previous section with the actual gradient updates. Following the kernel-based framework of LLM fine-tuning [18], the evolution of logits $\mathbf{z}$ is governed by the empirical neural tangent kernel (eNTK). We adopt the Relatively Stable Kernel (RSK) assumption [18], which posits that the relative interaction structure encoded by the token-level kernel $\mathbf{K}$ remains approximately stable during fine-tuning; this assumption is empirically supported in [18] and is consistent with prior observations (see Section 5).

Specifically, consider an update driven by a loss function with a gradient term $\mathbf{G}_k$ on logits $\mathbf{z}_k$ at step k . The change in logits at step l is $\Delta \mathbf{z}_l = -\eta \sum_k \mathbf{K}_{l,k} \mathbf{G}_k$. For Supervised Fine-tuning (SFT), the gradient corresponds to the prediction error $\mathbf{G}_k = \pi_k - \mathbf{p}_k$, where $\mathbf{p}_k \in \Delta^\mathcal{V}$ denotes the target next-token distribution. By projecting these logit updates onto our alignment decomposition, we arrive at the following general result:

Lemma 2.8 (General Alignment Evolution). *Under the RSK assumption, let $\mathbf{G}_l$ be a gradient term of the loss with respect to the logits at position l . The first-order change in the alignment score ΔS under the parameter update $\Delta \theta$ is given by:*

$$\Delta S = -\eta \mathbb{E}_{\mathbf{x} \sim p(\mathbf{x})} \sum_{m=1}^{L_y} \sum_{l=1}^{L_y} \pi_m^< \cdot \pi_{S,m}^+ \pi_{S,m}^- (\tilde{\pi}_m^+ - \tilde{\pi}_m^-)^\top \mathbf{K}_{m,l} \mathbf{G}_l + \mathcal{O}(\eta^2), \quad (6)$$

where $\pi_m^<$, $\pi_{S,m}^\pm$, and $\tilde{\pi}_m^\pm$ are defined at the m -th token position given the prefix $\mathbf{y}_{<m}$.

Lemma 2.8 indicates that alignment progress is determined by how the gradient field $\mathbf{G}$ aligns with the contrastive direction $\tilde{\pi}^+ - \tilde{\pi}^-$. Substituting the SFT gradient into this Lemma yields our main theoretical result:

Theorem 2.9 (SFT Alignment Dynamics). *Under the RSK assumption, the expected evolution of the alignment score during SFT over a data distribution $\mathcal{D}$ follows:*

$$\Delta \bar{S} = \eta \cdot \mathbb{E}_{(\mathbf{x}, \mathbf{y}) \sim \mathcal{D}} \left[\sum_{m=1}^{L_y} \pi_m^< \pi_{S,m}^+ \pi_{S,m}^- \cdot \left(\mathcal{F}_{\text{rebound}}^{(m)} + \mathcal{F}_{\text{drive}}^{(m)} \right) \right] + \mathcal{O}(\eta^2), \quad (7)$$

where for each position m , the forces are aggregated over all sequence positions l :

$$\mathcal{F}_{\text{rebound}}^{(m)} = \sum_{l=1}^{L_y} \left(-\pi_{S,m}^+ (\tilde{\pi}_m^+ - \tilde{\pi}_m^-)^\top \mathbf{K}_{m,l} \tilde{\pi}_l^+ + \pi_{S,m}^- (\tilde{\pi}_m^- - \tilde{\pi}_m^+)^\top \mathbf{K}_{m,l} \tilde{\pi}_l^- \right), \quad (8)$$

$$\mathcal{F}_{\text{drive}}^{(m)} = \sum_{l=1}^{L_y} (\tilde{\pi}_m^+ - \tilde{\pi}_m^-)^\top \mathbf{K}_{m,l} \mathbf{p}_l. \quad (9)$$

The decomposition in Theorem 2.9 reveals that the trajectory of alignment is governed by two competing forces:

Rebound Force ($\mathcal{F}_{\text{rebound}}$): representing the internal resistance or “distributional inertia”. It is a self-interaction term where the model’s current policy opposes further shifts in its distribution.

Driving Force ($\mathcal{F}_{\text{drive}}$): representing the external “pull” from the supervision signal. Geometrically, it is the projection of the data distribution $\mathbf{p}$ onto the alignment-contrastive direction, weighted by the model’s kernel $\mathbf{K}$.

Remark. To gain further insight, consider the identity kernel $\mathbf{K} = \mathbf{I}$. Here, the **Rebound Force** becomes $-\pi_S^+ \sum \tilde{\pi}_i^{+2} + \pi_S^- \sum \tilde{\pi}_j^{-2}$, where the sum of squares $\sum_i \pi_i^2$ measures the distributional purity (complementary to Gini impurity). Meanwhile, the **Driving Force** simplifies to a direct dot product $\langle \tilde{\pi}^+ - \tilde{\pi}^-, \mathbf{p} \rangle$. In the general case, $\mathbf{K}$ acts as a similarity-based weighting matrix, modulating these forces based on the similarity between samples [18, 20, 21].

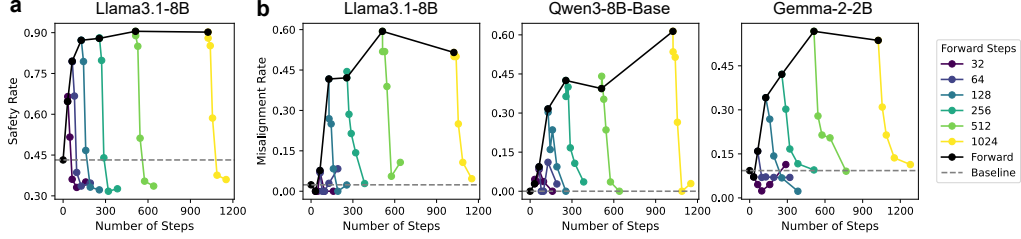


Figure 2: **Analysis of rebound dynamics.** (a) Results on the Beavertails dataset. (b) Results on the Emergent Misalignment dataset. Black lines represent the initial forward fine-tuning (Stage 1), while colored lines denote the subsequent reverse fine-tuning (Stage 2). The distinct colors indicate varying fine-tuning steps of Stage 1 training prior to the onset of Stage 2. We observe a rapid rebound to the baseline performance across all checkpoints, regardless of the number of forward fine-tuning steps.

3 Revisiting Rebound Effect: A Dynamical Perspective

The Rebound Effect [14] refers to the tendency of fine-tuned models to revert toward their pre-trained state under reversed or perturbed fine-tuning. Our dynamical framework (Theorem 2.9) provides a mechanistic view, modeling alignment as a competition between the external **Driving Force** $\mathcal{F}_{\text{drive}}$ and the internal **Rebound Force** $\mathcal{F}_{\text{rebound}}$.

We extend this phenomenon to an emergent misalignment dataset [22], showing that resistance to state shifts is bidirectional (Section 3.2). Our theory further predicts that the narrowness of the target distribution—captured by $\tilde{\pi}^{\pm\top} K \tilde{\pi}^{\pm}$ —modulates the rebound force, which we validate via controlled SFT experiments (Section 3.3).

3.1 General Experimental Setup

In this section, we describe the shared experimental configurations, models, and evaluation metrics used throughout our study.

The Multi-Stage SFT Paradigm. Our experiments follow a sequential Supervised Fine-Tuning (SFT) paradigm. In Stage 1 (Forward), the model is trained to reach a target alignment state (e.g., becoming safe). In Stage 2 (Reverse), we perturb this state by fine-tuning on a reversed or unrelated distribution to observe the Rebound Effect. We detail the dataset preparation in Appendix B.3.1. Section 4 further extends this to Stage 3 (Re-alignment), where the model is re-exposed to the Stage 1 distribution to study the Rehearsal Priming effect.

Evaluation Metrics. We quantify the alignment state S using three domain-specific metrics. **Safety Rate:** The percentage of safe responses on sampled malicious prompts, judged by ShieldGemma-9B [23]. **Misalignment Rate:** percentage of completions judged as misaligned by the ShieldGemma-9B rubric on 44 elicitation prompts [22]. **Positive Score:** The probability of positive sentiment in the generated text, as estimated by a Sentiment RoBERTa evaluator [24]. Refer to Appendix B.2.1 for complete implementation details.

Model Selection and Pre-processing. We evaluate our theory on three representative base LLMs: Llama-3.1-8B [25], Gemma-2-2B [26], and Qwen3-8B² [27]. We intentionally select base models rather than instruction-tuned versions to eliminate the confounding effects of prior alignment (e.g., RLHF). To ensure basic instruction-following capability, all models undergo an initial warm-up fine-tuning on the Alpaca-cleaned dataset [28] before proceeding to the alignment experiments.

3.2 Rebound Effect in Both Alignment and Misalignment Fine-Tuning

Experimental Configurations. To empirically validate the bidirectional nature of the rebound effect, we apply the two-stage SFT paradigm to both alignment and misalignment objectives. Specifically, for safety alignment, we transition the model from safety-aligned responses (distilled from Qwen3-8B)

²Since its release checkpoints already incorporate safety alignment, Qwen3-8B-Base is excluded from “safety rate” metrics but is still evaluated for positive score and misalignment rate.